Test report 12 – fixes for colors and daemon

Findings F1–F9 of the independent review (`tests/farben-pruefung/`) and daemon socket path · cad-eval, cad-engine, cad-session, cad-cli, cad-ipc, web · 2026-10-23 · Branch worktree-agent-a118501517e2aa7b5 · Commit 1968be9 · rustc 1.97.0 (2d8144b78 2026-07-07) · 4 cores

cargo test --workspace	clippy · fmt	Vitest · Playwright	Tauri-E2E	OpenSCAD · STEP
404 / 404	**clean · clean**	**195 · 88 / 88**	**15 / 15**	**33/33 · 48/48**
of which 16 new regression tests	-D warnings	new: G01–G04	new: T15	compare.py · verify.py

Everyday models	3MF (lib3mf, trimesh)
unchanged	**passed**
tests/alltag/check.sh	Bambu color dialog: no

1 Task and result

An independent test agent reviewed the color features from report 11 and reported nine findings (F1–F9) and one daemon bug (`tests/farben-pruefung/README.md`, test cases in `cases/`, images in `img/`). All findings are fixed; each has a regression test in Rust, the UI findings additionally in Vitest, Playwright and Tauri E2E. The review's test cases all run as `crates/cad-cli/tests/farben_pruefung.rs` (each file: no errors, region volume = part volume, random colors at a distance from the palette). "Before" images are the reviewer's (commit d8fe983), "after" images are from the CLI and UI of this commit.

No.	Severity	Finding	Status
F1	high	Same geometry in different colors: view and STEP in the first color	fixed
F2	high	Slot choice in the UI on a color() with a parameter/loop color affected all colors	fixed (filament() table)
F3	medium	View ≠ print (overlap, fillet at a color boundary, color() in the 2D profile)	fixed (regions in the view; mixed 2D colors: warning)
F4	medium	Material outside parts: UI omitted it, CLI message wrong	fixed
F5	medium	Random colors collide	fixed
F6	medium/low	Slivers between neighbors of the same color	fixed
F7	low	Alpha variants, silent clamping, transparent, empty name	fixed (rules + warnings)
F8	low	> 16 slots, shared slots without notice, color group for single-color	fixed
F9	low	--per-color with .stl, color list with --part, file name with use/include	fixed
D	–	Daemon silently fails to start with a socket path that is too long	fixed (substitute path)
B	–	Bambu Studio: color dialog overwrites slots	avoided (3MF without color group)

2 Findings

F1 (high) – same geometry, different colors

Cause The cache key of a node does not contain the color (intended: a new color recomputes nothing). `color("red") cube(5)` and `translate(…) color("blue") cube(5)` therefore have the same cube node

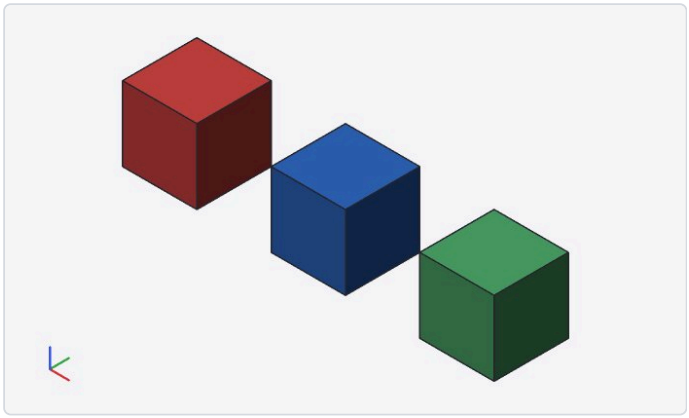
content, the same face tag and – from the cache – the same kernel faces. The view (`tagColors`: tag → first node with color) and STEP (`nodes_for_tag(tag).first()`) took the first color for all faces of this tag. The 3MF was correct because the color regions place the instances individually.

Fix Cache reuse stays; the color is assigned via instances and regions: (1) for models with colors the view shows the **color regions** (the bodies of the 3MF) as soon as they are computed (UI "Show color regions" (German UI: „Farbregionen zeigen“), now the default; `cad render` without `--show faces`); (2) the color per face (before that, for highlighted subtrees and `--show faces`) only takes nodes below the drawn node and, among several with the same content, the one whose bounding box contains the triangle (`faceColors` in `web/src/logic/colors.ts`, `face_colors` in `cad-cli/src/render.rs`); (3) STEP face colors come from the rule set of the regions without boolean: sample points of each face against the placed material of each color node (`kernel.classify`), the latest wins (`Engine::face_region_colors`), as synthetic tags in a copy of the B-rep.

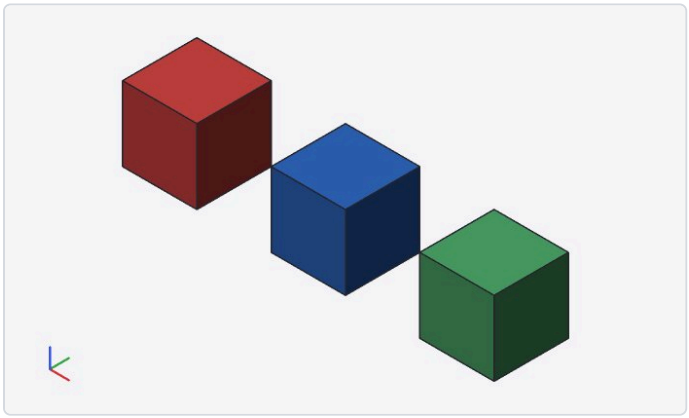
Tests `f1_equal_geometry_keeps_its_colour_in_view_and_step` **ok**, `all_review_cases_evaluate_and_regions_add_up` **ok**; Vitest „Bericht 12 F1“; Playwright G01; Tauri T15.



before: pal2.scad, three red cubes (reviewer)



after: `cad render` (color regions)



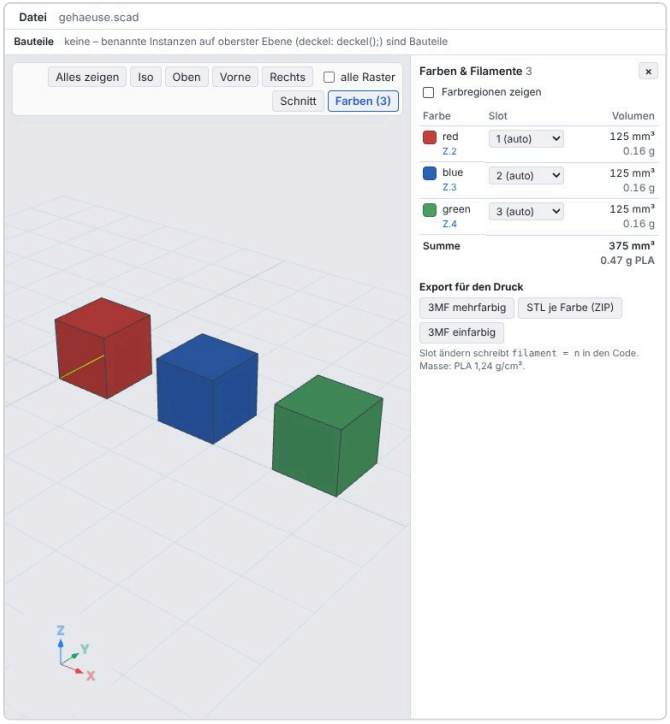
after: `--show faces` (color per face per instance)



after: palette.scad, 16 identical tiles in a loop

STEP of pal2.scad (faces per color):

```
#2e65b9: 6 Flächen
#4ca167: 6 Flächen
#c74240: 6 Flächen
```



UI: color per face with "Show color regions" off – each instance in its color

Limitation: the color per face is an approximation (bounding box test; with overlapping bounding boxes of identically shaped bodies of different color, the later one). Exact are the regions that the view shows after the computation; in STEP a face merged across a color boundary gets one color (majority of the sample points).

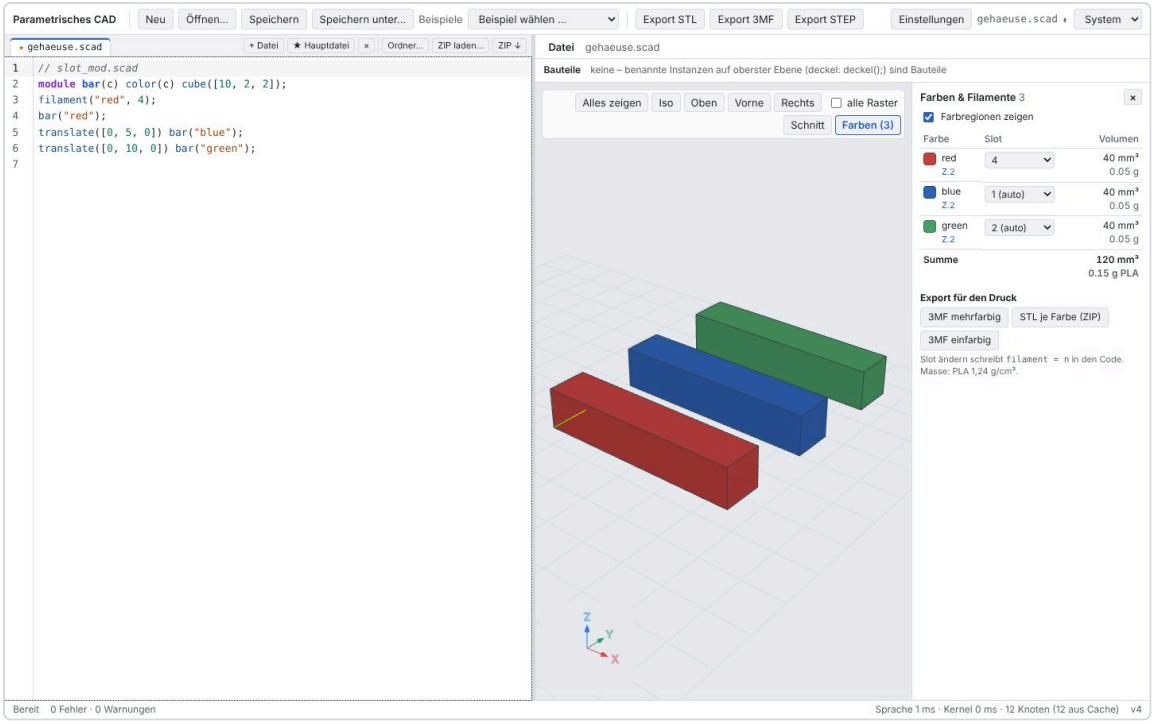
F2 (high) – slot choice with a parameter or loop color

Cause `SetFilament` wrote `filament = n` on the first `color()` statement of the color. With `module bar(c) color(c) …` or `for (i) color(names[i]) …` this statement colors several colors – all of them got the slot (or the first specification won with a warning).

Decision A slot table in the code: `filament("red", 4);` at the top level of the main file, one statement per color. Reason: the code stays the only source (CLI and UI export the same, undo/Git see the change, an agent reads it too); the table applies to exactly one color and is independent of where the color arises (parameter, loop, library); it is an ordinary module call (no new syntax, OpenSCAD only reports an unknown module); a block `filaments { … }` would need parser changes, `$filaments = […]` would have the scoping rules of `$` variables and could be overridden anywhere. The UI still writes `filament = n` on the statement if **all** statements of the color name it as a literal (readable in place), otherwise the table line; if there is a table line, it changes its slot, "automatic" removes the line.

Rules The table is evaluated before all other statements (separate pass) and takes precedence; a contradicting `filament = m` warns and is ignored; `filament()` in modules, loops or libraries warns („nur auf oberster Ebene der Hauptdatei“, "only at the top level of the main file"); a user's own module `filament` takes precedence.

Tests `filament_declarations` **ok**, `filament_table_rules` **ok**, `set_filament_of_a_parameter_colour_writes_a_table` **ok**; Playwright G02; Tauri T15.



UI after slot 4 for red in slot_mod.scad: `filament("red", 4);` in the code, blue and green automatic (before, all three got slot 4)

`cad outline` of the file with table:

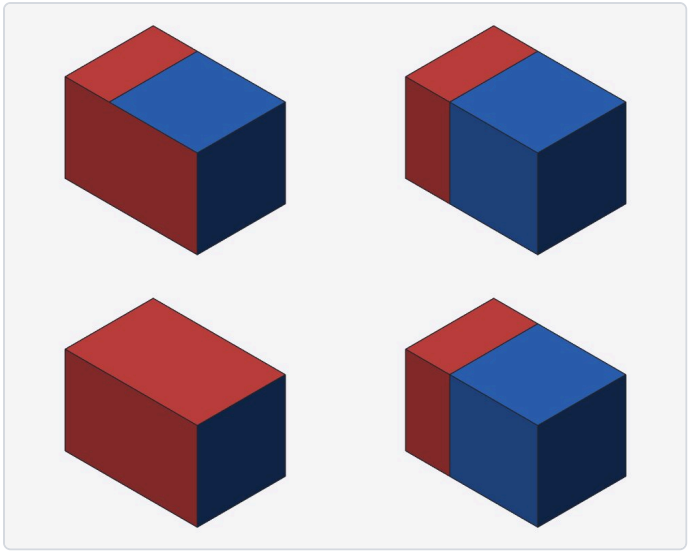
```
Farben (Filament, Farbe, Volumen, Masse PLA 1,24 g/cm³):
  F4  red            #c74240  40 mm³  0.05 g  Z.1  filament = 4
  F1  blue           #2e65b9  40 mm³  0.05 g  Z.1
  F2  green          #4ca167  40 mm³  0.05 g  Z.1
```

F3 (medium) – view ≠ print

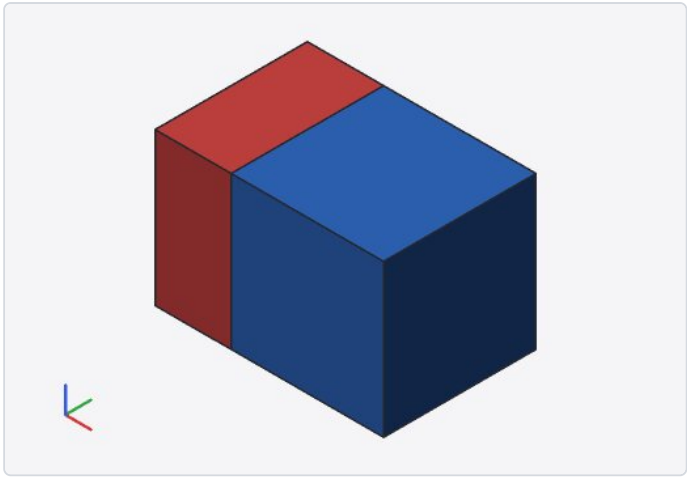
Cause (a) With overlap the kernel merges coplanar faces of both bodies into one face with one tag – the color per face cannot show the color boundary. (b) The fillet at a color boundary belongs to the uncolored `union` node (shown gray), the regions give it the remainder color (red). (c) `color()` in the 2D profile: the view showed the color of the 2D leaf, the color table did not look below profile operations (default, no entry, no warning).

Fix (a, b) The view shows the color regions (F1). (c) If all 2D leaves of an extrusion without its own color have the same color, the extrusion is a body of that color (as with `color()` around it); mixed 2D colors are not separated: one body in the default color and the warning „Farben im 2D-Profil von linear_extrude werden nicht getrennt …“ ("colors in the 2D profile of linear_extrude are not separated …").

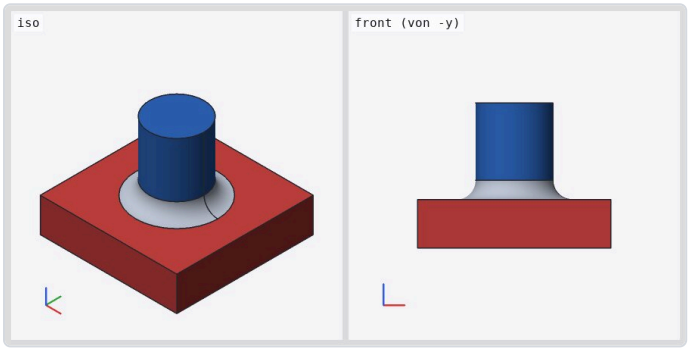
Tests `f3_view_matches_the_print` **ok**; Playwright G03.



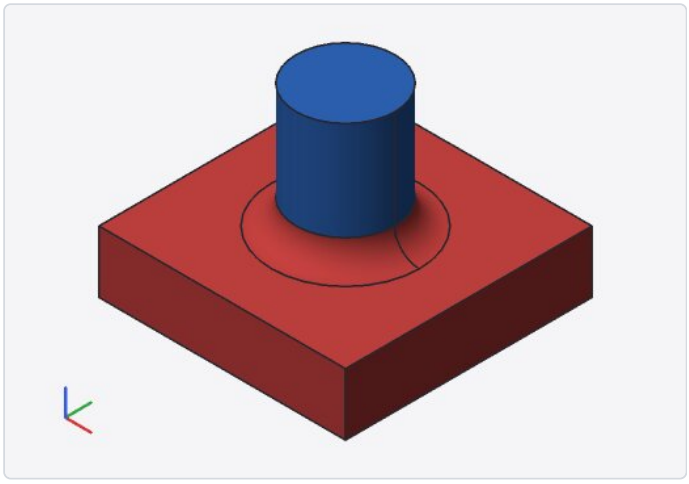
before: r_overlap2/3 (left), regions (right) – reviewer



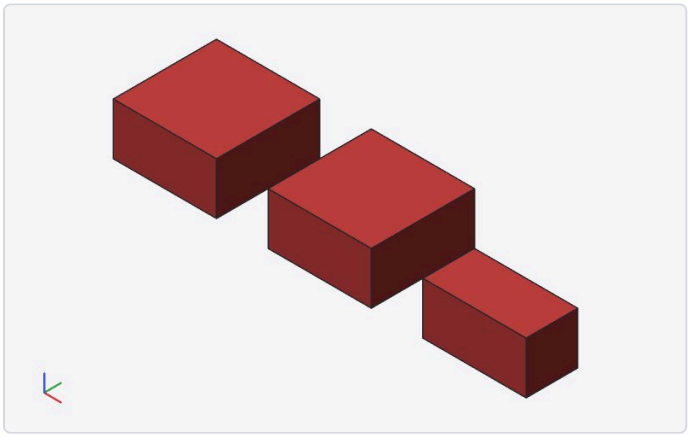
after: r_overlap3.scad, blue owns the overlap



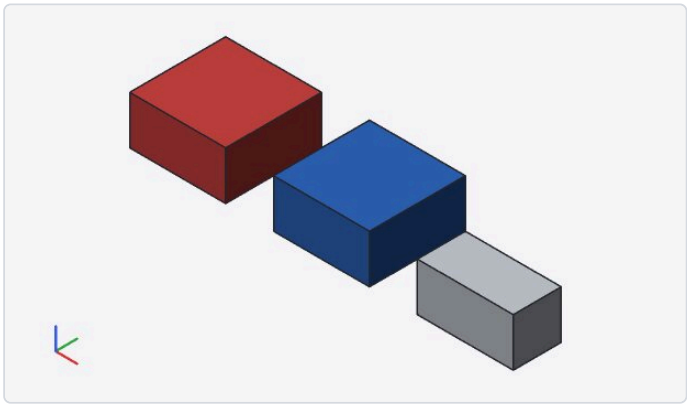
before: fillet gray, exported red (reviewer)



after: fillet red as in the print



before: 2D color shown, exported as default (reviewer)



after: red and blue as exported, mixed profile default with warning

```
tests/farben-pruefung/cases/r_extrude.scad:3:21: Warnung (Auswertung): Farben im 2D-Profil von linear_extrude
werden nicht getrennt: der Körper druckt als Standard – color() um die Extrusion setzen oder je Farbe
extrudieren
Farben (Filament, Farbe, Volumen, Masse PLA 1,24 g/cm³):
  F1  Standard       #c3c8cf  250 mm³  0.31 g  (ohne color())
  F2  red            #c74240  500 mm³  0.62 g  Z.1
  F3  blue           #2e65b9  500 mm³  0.62 g  Z.2
```

Limitation: mixed colors in a 2D profile below an extrusion are not split into bodies per color (warning; remedy: extrude per color).

F4 (medium) – material outside parts

Cause Color list and UI export only took the parts (rest: 0 mm³, omitted in the 3MF); the CLI export without `--part` took the root as one object and still reported "one object per part".

Fix The regions comprise the parts and the top-level nodes that are neither a part nor contain one („ohne Bauteil“, "without part", `cad_session::rest_nodes`); the default export (CLI without `--part` and UI, both `Session::export_colors`) writes one object per part plus the object „ohne Bauteil“ – the files from CLI and UI are identical. The message lists the objects (`color_export_notes`), the UI shows them in the status line.

Tests `f4_material_outside_parts` **ok**, `material_outside_parts_is_exported` **ok**; Playwright G04; Tauri T15.

```
tests/farben-pruefung/cases/inc/lib.scad:2:22: Warnung (Auswertung): Farbe `rosarot` unbekannt – Zufallsfarbe
(Palette: red, orange, yellow, lime, green, teal, cyan, blue, purple, magenta, pink, brown, black, gray,
silver, white)
main_use.3mf: 792 Dreiecke, 55988 Bytes, Volumen 1204.3893 mm³
3MF mehrfarbig: 3 Objekte (teil, deko, ohne Bauteil), 6 Teile (eines je Farbe und Objekt); Material außerhalb
der Bauteile ist ein eigenes Objekt „ohne Bauteil“; Filament-Slots in Metadata/model_settings.config, ohne
Farbgruppe: Bambu Studio und OrcaSlicer übernehmen die Slots ohne Farbdialog (einfarbig: --single-color)
Farben (Filament, Farbe, Volumen, Masse PLA 1,24 g/cm³):
  F1  white          #f4f3f0  600 mm³  0.74 g  Z.2
  F2  red            #c74240  150.7964 mm³  0.19 g  lib.scad:1
  F3  teal           #217274  301.5929 mm³  0.37 g  lib.scad:1
  F4  rosarot        #959d3b  27 mm³  0.03 g  lib.scad:2  (unbekannter Name: Zufallsfarbe)
  F5  yellow         #f7d974  125 mm³  0.16 g  Z.6
  teil: F1 white 600 mm³, F2 red 150.7964 mm³, F3 teal 150.7964 mm³, F4 rosarot 27 mm³
  deko: F3 teal 150.7964 mm³
  ohne Bauteil: F5 yellow 125 mm³
```

F5 (medium) – random colors collide

Cause Only the hue came from the hash (L, C fixed): names fell onto the same value (`baz`/`violet`, `x`/`rose`) and were merged into one entry and slot; others lay close to palette colors.

Fix Hue, lightness (0.45–0.80) and chroma (0.06–0.16) from the hash; one candidate sequence per name (SplitMix64); the first candidate with $\Delta E_{OK} \geq 0.07$ to palette and default and – within the model, in order of first occurrence – ≥ 0.06 to the random colors of other names is chosen. Different names thus never coincide by chance; entries are merged only for exactly the same value (e.g. `"#f00"` and `[1, 0, 0]`).

Evidence randnames.scad: 34 unknown names (36 statements, rot/Rot/ROT is one name), 34 entries on 34 different slots; smallest distance between two random colors ΔE_{OK} 0.060 (gelb/violet), to the palette 0.071 (rot ↔ #c74240).

Tests `f5_random_colours_keep_their_distance` **ok**, `random_colours_do_not_collide_in_a_model` **ok**.

rot · grün · blau · gelb · lightblue · darkred · navy · olive · maroon · coral · salmon · gold · violet · indigo · beige · tan · khaki · crimson · turquoise · lavender · mint · rose · foo · bar · baz · x · y · z · petrol · schwarz · weiss · orange2 · red2 · blue1

Limitation: if a name has to move away within the model, its color depends on the names before it (deterministic, but not the same across models).

F6 – slivers between neighbors of the same color

Cause `Stück(C) = R ∩ M(C) − ⋃ M(später)` also subtracted a touching later body of the same color; thin slivers (0.00016 mm) remained at the contact face, which overlapped after the union: r_touch.scad red 2000.032 instead of 2000 mm³, without warning.

Fix Only the material of later color nodes of a **different** color is subtracted; the pieces of one color are united as before. The assignment is the same (same color), the slivers disappear.

Tests `f6_no_slivers_between_same_coloured_neighbours` **ok**, `all_review_cases_evaluate_and_regions_add_up` **ok** (all cases: sum = volume to 10^{-6}).

```
Farben (Filament, Farbe, Volumen, Masse PLA 1,24 g/cm³):
  F1  red           #c74240  2000 mm³  2.48 g  Z.1, Z.2
  F2  blue          #2e65b9  1000 mm³  1.24 g  Z.3
```

F7 – alpha, clamping, transparent, empty name

Rules The identity of a print color is its RGB value: `alpha` only affects the view (the color region is drawn transparent), `color("red", alpha = 0.5)` is the entry `red` (one slot, the conflict check ignores alpha). Vector values and `alpha` outside 0–1 are clamped with a warning (for 0–255 values with a hint to `"#rrggbb"`). `color("transparent")` is not a print color: warning, the material stays default (no filament `#00000000`); `color("")` and `color()` warn and do not color.

Tests `f7_value_rules` **ok**.

```
tests/farben-pruefung/cases/values.scad:8:27: Warnung (Auswertung): Farbwerte liegen zwischen 0 und 1 (wie in
OpenSCAD), andere werden geklemmt – für Werte 0–255 "#ff0000" schreiben
tests/farben-pruefung/cases/values.scad:9:27: Warnung (Auswertung): Farbe muss Name oder [r, g, b, a] sein
tests/farben-pruefung/cases/values.scad:10:27: Warnung (Auswertung): Farbe muss Name oder [r, g, b, a] sein
tests/farben-pruefung/cases/values.scad:11:27: Warnung (Auswertung): `#12345` ist keine Farbe (#rgb, #rrggbb,
#rrggbbaa oder Name)
tests/farben-pruefung/cases/values.scad:12:27: Warnung (Auswertung): `#ggg` ist keine Farbe (#rgb, #rrggbb,
#rrggbbaa oder Name)
tests/farben-pruefung/cases/values.scad:13:27: Warnung (Auswertung): Farbe muss Name oder [r, g, b, a] sein
tests/farben-pruefung/cases/values.scad:14:42: Warnung (Auswertung): alpha liegt zwischen 0 und 1, der Wert
wird geklemmt
tests/farben-pruefung/cases/values.scad:15:27: Warnung (Auswertung): `transparent` ist keine Druckfarbe: das
Material bleibt ungefärbt (Standard, Filament 1)
tests/farben-pruefung/cases/values.scad:18:27: Warnung (Auswertung): Farbe muss Name oder [r, g, b, a] sein
tests/farben-pruefung/cases/values.scad:19:27: Warnung (Auswertung): Farbwerte liegen zwischen 0 und 1 (wie in
OpenSCAD), andere werden geklemmt
tests/farben-pruefung/cases/values.scad:20:42: Warnung (Auswertung): alpha liegt zwischen 0 und 1, der Wert
wird geklemmt
tests/farben-pruefung/cases/values.scad:21:42: Warnung (Auswertung): `alpha` muss eine Zahl sein, ist String
tests/farben-pruefung/cases/values.scad:22:34: Warnung (Auswertung): `color()` hat keinen Parameter `foo`
tests/farben-pruefung/cases/values.scad:23:21: Warnung (Auswertung): color() ohne Farbe hat keine Wirkung
Farben (Filament, Farbe, Volumen, Masse PLA 1,24 g/cm³):
  F1  Standard      #c3c8cf  81.845 mm³  0.1 g  (ohne color())
  F2  #ff0000       #ff0000  8.11 mm³  0.01 g  Z.1, Z.4, Z.8
  F3  #00ff00       #00ff00  1.331 mm³  0 g  Z.2
  F4  #0000ff       #0000ff  1.728 mm³  0 g  Z.3
  F5  #80331a       #80331a  6.119 mm³  0.01 g  Z.5, Z.6
  F6  teal          #217274  4.096 mm³  0.01 g  Z.7
  F7  red           #c74240  94.354 mm³  0.12 g  Z.14, Z.17, Z.20, Z.21, Z.22
  F8  blue          #2e65b9  12.5 mm³  0.02 g  Z.16
  F9  #008080       #008080  15.68 mm³  0.02 g  Z.19
```

F8 – slots above 16, shared slots, single-color 3MF

Fix The color table reports automatic slots above 16 as a warning („mehr als 16 Filamente …“, "more than 16 filaments …"), two colors on the same fixed slot and a fixed slot 1 next to the default as a notice (info). The 3MF no longer has a color group (section 3), not even with `--color-group` when there is only one color.

Tests `f8_slot_remarks_and_single_colour` **ok**, `color_export::tests::no_color_group_by_default` **ok**.

```
tests/farben-pruefung/cases/f1.scad:3:45: Warnung (Auswertung): Farbe hat schon Filament 3 (die erste Angabe
gilt), filament = 4 wird ignoriert
tests/farben-pruefung/cases/f1.scad:8:47: Warnung (Auswertung): filament muss eine ganze Zahl von 1 bis 16 sein
tests/farben-pruefung/cases/f1.scad:9:47: Warnung (Auswertung): filament muss eine ganze Zahl von 1 bis 16 sein
tests/farben-pruefung/cases/f1.scad:10:47: Warnung (Auswertung): filament muss eine ganze Zahl von 1 bis 16
sein
tests/farben-pruefung/cases/f1.scad:11:48: Warnung (Auswertung): `filament` muss eine Zahl sein, ist String
tests/farben-pruefung/cases/f1.scad:13:46: Warnung (Auswertung): Farbe hat schon Filament 3 (die erste Angabe
gilt), filament = 5 wird ignoriert
tests/farben-pruefung/cases/f1.scad:14:50: Warnung (Auswertung): Farbe hat schon Filament 3 (die erste Angabe
gilt), filament = 6 wird ignoriert
tests/farben-pruefung/cases/f1.scad:15:59: Warnung (Auswertung): Farbe hat schon Filament 3 (die erste Angabe
gilt), filament = 7 wird ignoriert
tests/farben-pruefung/cases/f1.scad:4:21: Info (Auswertung): Slot 3 hat schon die Farbe `red`: beide drucken
mit demselben Filament
tests/farben-pruefung/cases/f1.scad:6:21: Info (Auswertung): Slot 1 druckt auch das Material ohne color()
(„Standard“) – `yellow` und Standard teilen sich das Filament
Farben (Filament, Farbe, Volumen, Masse PLA 1,24 g/cm³):
  F1  Standard       #c3c8cf  125 mm³  0.16 g  (ohne color())
  F3  red            #c74240  750 mm³  0.93 g  Z.1, Z.2, Z.3, Z.13, Z.14, Z.15  filament = 3
  F3  blue           #2e65b9  125 mm³  0.16 g  Z.4  filament = 3
  F2  green          #4ca167  125 mm³  0.16 g  Z.5
  F1  yellow         #f7d974  125 mm³  0.16 g  Z.6  filament = 1
  F4  white          #f4f3f0  500 mm³  0.62 g  Z.8, Z.9, Z.10, Z.11
  F16 pink           #f7b8c6  125 mm³  0.16 g  Z.12  filament = 16
```

F9 – --per-color, --part, use/include

Fix `--per-color` requires the extension `.zip` (otherwise exit 2 with a suggestion); `--per-color --part` shows only the colors and volumes of the selected parts; the color list names lines of other files as `lib.scad:1` (CLI and UI; in the UI a click opens the file at that position).

Tests `f9_per_colour_export_and_list` **ok**; Vitest „Bericht 12 F9“.

```
cad export: --per-color schreibt eine ZIP-Datei mit einem STL je Farbe: -o x.zip

tests/farben-pruefung/cases/inc/lib.scad:2:22: Warnung (Auswertung): Farbe `rosarot` unbekannt – Zufallsfarbe
(Palette: red, orange, yellow, lime, green, teal, cyan, blue, purple, magenta, pink, brown, black, gray,
silver, white)
teil.zip: 4 STL (eine je Farbe), 528 Dreiecke, 27156 Bytes
Farben (Filament, Farbe, Volumen, Masse PLA 1,24 g/cm³):
  F1  white          #f4f3f0  600 mm³  0.74 g  Z.2
  F2  red            #c74240  150.7964 mm³  0.19 g  lib.scad:1
  F3  teal           #217274  150.7964 mm³  0.19 g  lib.scad:1
  F4  rosarot        #959d3b  27 mm³  0.03 g  lib.scad:2  (unbekannter Name: Zufallsfarbe)
```

Daemon – socket path longer than the Unix limit

Cause `sun_path` holds 108 bytes (Linux; 107 without NUL). With a deep `CAD_IPC_DIR`, `bind` failed ("path must be shorter than SUN_LEN") and the client silently computed without the daemon.

Fix `cad_ipc::socket_path_in`: if `<dir>/<name>.sock` is too long, the socket is placed in `/tmp/cad-<uid>/<hash>-<name>.sock` (directory with the same checks: 0700, own user, no symlink; `hash` = FNV-1a of the canonical runtime directory, so that several long directories do not interfere). Lock and log file stay in the runtime directory; `list` also finds substitute sockets (UI endpoints); `cad daemon status` shows the path with „Ersatzpfad“ ("substitute path", JSON `socket_substitute`), the log names it.

Tests `long_ipc_dir_uses_short_socket_path` **ok**.

```
CAD_IPC_DIR: /tmp/tmp.OVFKN13EtA/ein-sehr-langer-verzeichnisname-fuer-den-socket-des-hintergrundprozesses-
damit-es-reicht/ipc (124 Byte bis daemon.sock)
cad: Daemon gestartet (pid 28107, Log /tmp/tmp.OVFKN13EtA/ein-sehr-langer-verzeichnisname-fuer-den-socket-des-
hintergrundprozesses-damit-es-reicht/ipc/daemon.log)
cad: Daemon pid 28107: 529 ms, Cache 13 Treffer / 64 neu, 64 Einträge, 7,8 MiB
Daemon läuft: pid 28107, Version 0.1.0 (Build 0.1.0+1ebe850.18e1397934aa36b8)
Laufzeit 0 s, Leerlauf 0 s (Ende nach 15 min ohne Anfrage), Watchdog 60 s
Anfragen: 1, Projekte 1 von 8 (0 verdrängt)
Cache: 62 Knoten + 2 Netze/abgeleitete, 7,8 MiB von 512,0 MiB, Trefferquote 17 % (13 Treffer, 64 neu
berechnet), 0 verdrängt
Speicher (RSS): 54,0 MiB, Start 36,1 MiB, Cache wird geleert über 804,1 MiB (0× geleert)
Socket /tmp/cad-0/27d121efb38cc86e-daemon.sock (Ersatzpfad: Pfad im IPC-Verzeichnis länger als 107 Byte), Log
/tmp/tmp.OVFKN13EtA/ein-sehr-langer-verzeichnisname-fuer-den-socket-des-hintergrundprozesses-damit-es-
reicht/ipc/daemon.log
Daemon beendet (pid 28107)
```

3 Bambu Studio: color dialog and slots

Sources retrieved on 2026-10-23 (`raw.githubusercontent.com`, BambuStudio `master`, OrcaSlicer `main`; checksums in `reports/tmp/report12/sources.sha256`):

```
== BambuStudio Plater.cpp: Farbdialog für fremde 3MF
8890:                    if (load_model && !color_group_map.empty() && !volume_color_data.empty() &&
en_3mf_file_type != En3mfType::From_BBS) {
== BambuStudio bbs_3mf.cpp: Dreiecksfarben nur aus pid/pindex
5517:            if (!m_is_bbl_3mf && sub_object->geometry.triangle_colors.size() == triangles_count) {
4328:            if (boost::starts_with(m_curr_characters, "BambuStudio-")) {
== BambuStudio bbs_3mf.cpp: model_settings.config für alle Dateien
1744:                else if (boost::algorithm::iequals(name, BBS_MODEL_CONFIG_FILE)) {
2130:                else if (boost::algorithm::iequals(name, BBS_MODEL_CONFIG_FILE)) {
== BambuStudio ObjColorDialog.cpp: OK überschreibt extruder
270:             m_panel_ObjColor->apply_volume_filament_ids();
647:void ObjColorPanel::apply_volume_filament_ids()
707:                    volume->config.set("extruder", volume_filament->second);
== OrcaSlicer bbs_3mf.cpp: Farbgruppe nur ohne Volumen-Metadaten
2307:                    auto extruder_itor = color_group_id_to_extruder_id_map.find(current_object-
>second.pid);
== OrcaSlicer Plater.cpp: Vorkommen von color_group_map (0: kein Farbdialog für 3MF)
0
```

- **Condition of the dialog** (`Plater.cpp`): `load_model && !color_group_map.empty() && !volume_color_data.empty() && en_3mf_file_type != From_BBS`. The importer fills `volume_color_data` only for sub-parts whose triangles have a color via `pid`/`pindex` (`bbs_3mf.cpp`, `triangle_colors.size() == triangles_count`).
- **OK in the dialog** calls `apply_volume_filament_ids` and resets `extruder` of every sub-part; "Cancel" (German UI: „Abbrechen“) and closing leave it.
- **`Metadata/model_settings.config`** is read by the importer for all 3MF files (not tied to the version check), and it takes over `extruder` per sub-part and object.
- **Ways to avoid the dialog:** (1) no color group – chosen; (2) `Application = BambuStudio-…` – rejected: claims a foreign vendor, switches on the importer's project paths (version check, expected project files, plates) and could break with new Bambu versions; (3) `project_settings.config` with `filament_colour` – rejected: not loaded for foreign files (`dont_load_config` without Bambu version), OrcaSlicer would apply it over printer and process settings; (4) paint data (`mmu_segmentation`) – unnecessary, the sub-parts already carry the slot.
- **OrcaSlicer** does not have this dialog; its importer uses color groups only for objects without volume metadata (`color_group_id_to_extruder_id_map`), with `model_settings.config` the `extruder` values apply. Dropping the color group is therefore compatible with Orca.

Implementation: the multi-color 3MF no longer has `m:colorgroup` nor `pid`/`pindex`; the slots are only in `model_settings.config`. In Bambu Studio the colors of the slots come from the printer's filaments (AMS). For programs that display 3MF colors, `cad export --color-group` still writes the color group; the message then says

„Bambu Studio öffnet einen Farbdialog – dort ‚Abbrechen' wählen" ("Bambu Studio opens a color dialog – choose Cancel there").

File	lib3mf	trimesh	Bambu model: extruder per sub-part	Color dialog (model of Plater.cpp)
mehrfarbig.3mf (default)	ok	watertight	schild: 1/2/3, figur: 4/1/5	does not open
mehrfarbig-cg.3mf (`--color-group`)	ok	watertight	schild: 1/2/3, figur: 4/1/5	opens ("Cancel" keeps slots)

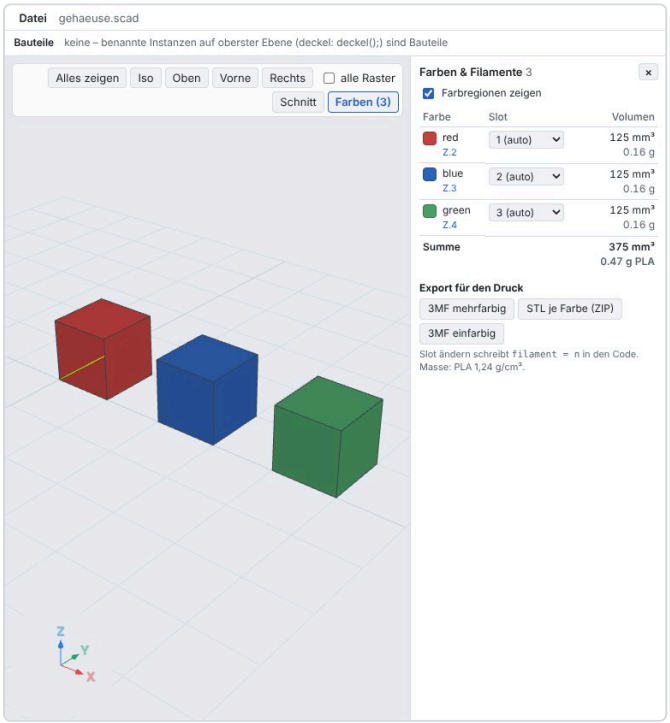
```
mehrfarbig.3mf: 8434 Dreiecke, 605379 Bytes, Volumen 7647.6695 mm³
3MF mehrfarbig: 2 Objekte (schild, figur), 6 Teile (eines je Farbe und Objekt); Filament-Slots in
Metadata/model_settings.config, ohne Farbgruppe: Bambu Studio und OrcaSlicer übernehmen die Slots ohne
Farbdialog (einfarbig: --single-color)
Farben (Filament, Farbe, Volumen, Masse PLA 1,24 g/cm³):
  F1  white          #f4f3f0  4900.8071 mm³  6.08 g  Z.36, Z.66
  F2  teal           #217274  622.5 mm³  0.77 g  Z.37
  F3  red            #c74240  208.8 mm³  0.26 g  Z.42  filament = 3
  F4  blue           #2e65b9  1689.3253 mm³  2.09 g  Z.61
  F5  yellow         #f7d974  226.2372 mm³  0.28 g  Z.67
  schild: F1 white 4773.5726 mm³, F2 teal 622.5 mm³, F3 red 208.8 mm³
  figur: F4 blue 1689.3253 mm³, F1 white 127.2345 mm³, F5 yellow 226.2372 mm³
```

Limitation: not checked in the programs themselves (Bambu Studio, OrcaSlicer) – no graphical program in the container. Whether Bambu Studio adds missing filaments on loading when a sub-part uses slot 5 and the project has only four filaments depends on the loading path (`Plater.cpp`: with "geometry only", filaments up to the highest number are created).

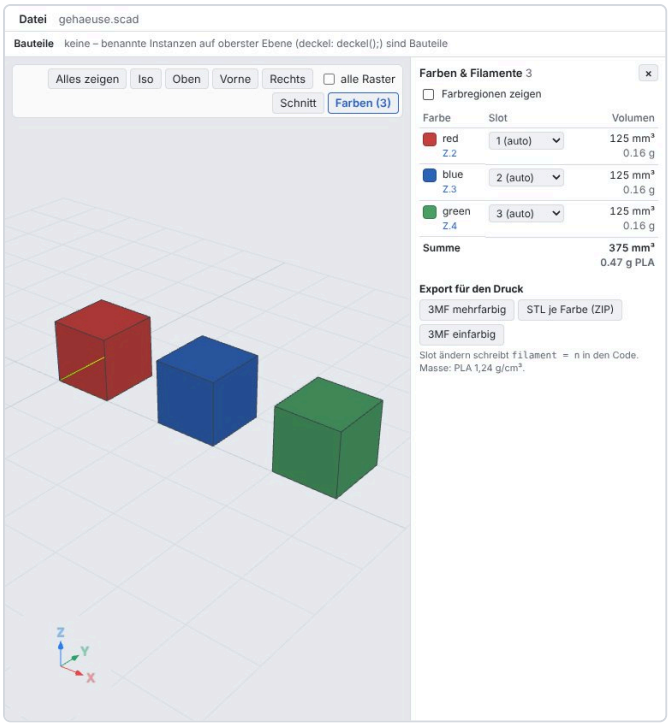
4 UI and desktop

G01 F1: same geometry in different colors

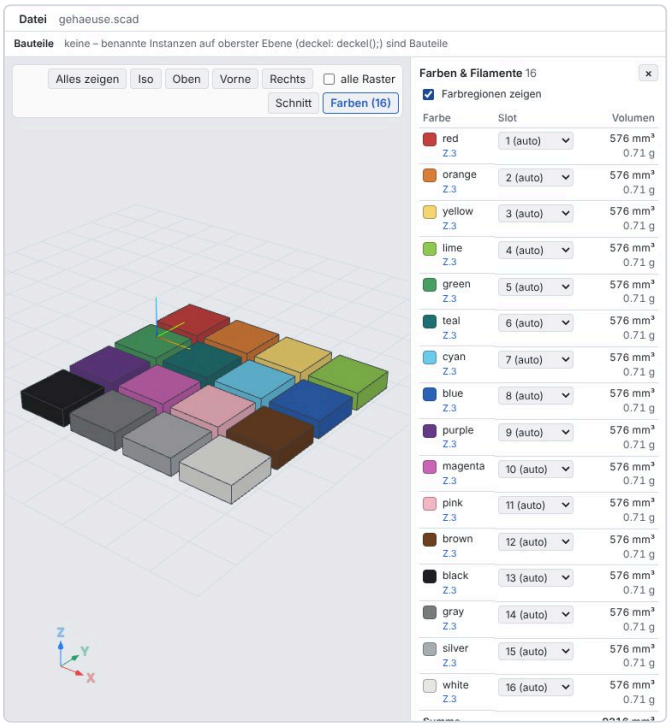
pal2.scad (three identical cubes red, blue, green) and palette.scad (16 identical tiles in a loop): before, all in the first color. Now the view shows the color regions (the bodies of the 3MF), and the color per face also takes the color of its own instance.



pal2: color regions (default)



pal2: color per face (regions off)

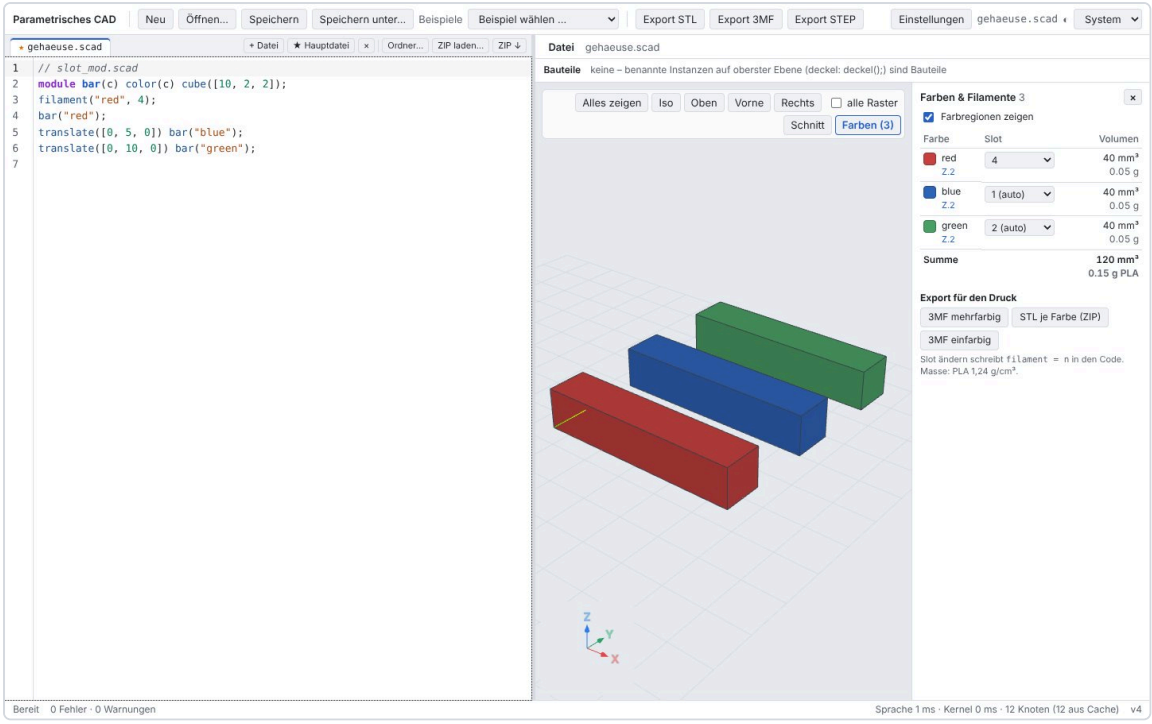


palette.scad: 16 tiles

- regions: 9960 red, 11988 blue, 13119 green pixels
- per face: 9960 red, 11988 blue, 13119 green pixels
- palette: 2381 teal, 2939 magenta pixels, 16 regions

G02 F2: slot of a color from a parameter

slot_mod.scad: module bar(c) color(c) ...; slot 4 for red previously wrote filament = 4 on color(c) – all three colors got slot 4. Now the UI writes filament("red", 4); into the file; blue and green stay automatic. "automatic" removes the line again.

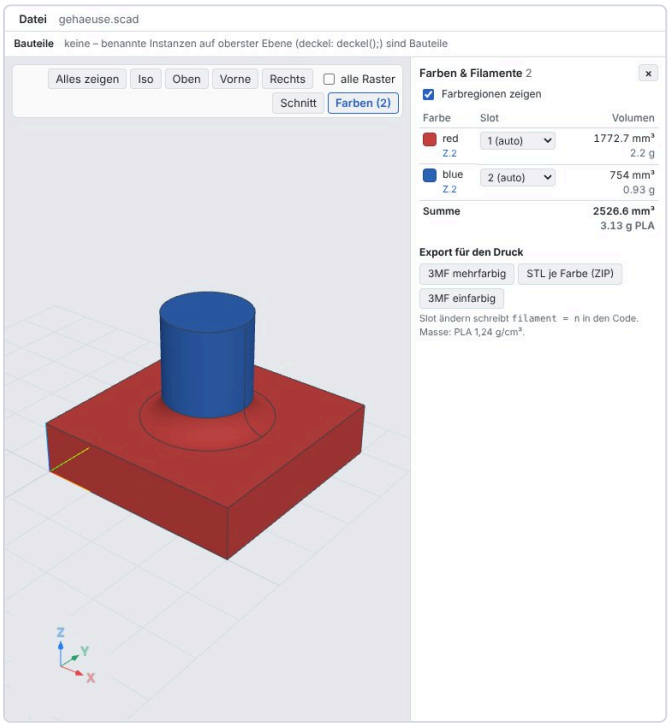


filament("red", 4); in the code, color list

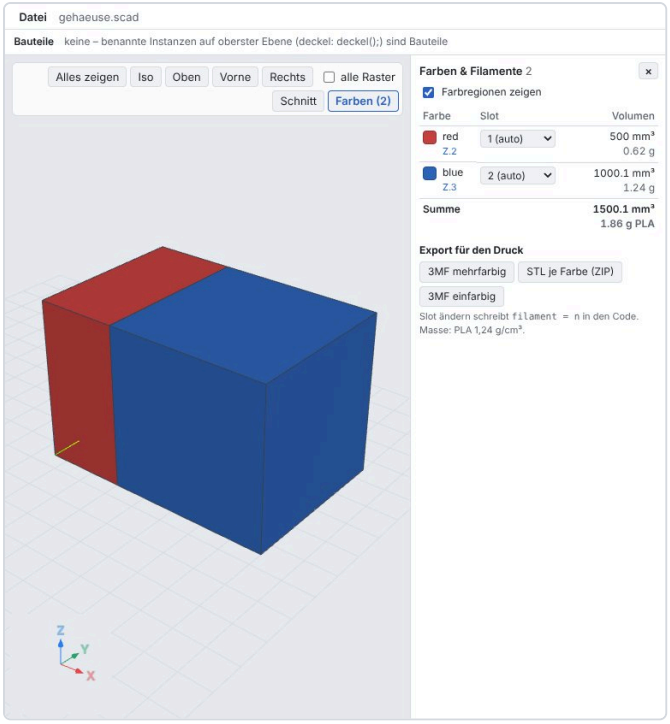
- text contains filament("red", 4); and no filament = 4
- slots red 4 (fixed), blue 1, green 2
- automatic: text as before

G03 F3: view = print

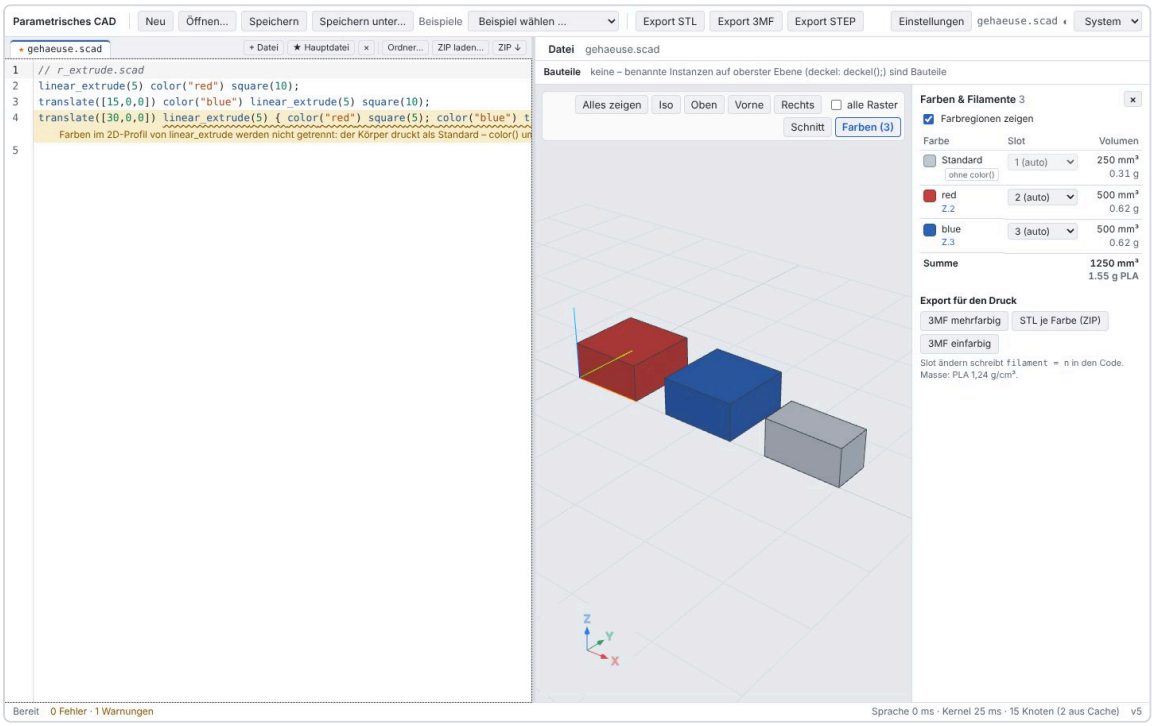
r_fillet2.scad: the fillet between red and blue was shown gray and exported red, now shown red. r_overlap3.scad: the later color (blue) owns the overlap, also in the view. r_extrude.scad: color() in the 2D profile colors the extrusion (red, blue); mixed 2D colors below an extrusion are not separated – warning, default.



Fillet at the color boundary



Overlap: blue wins

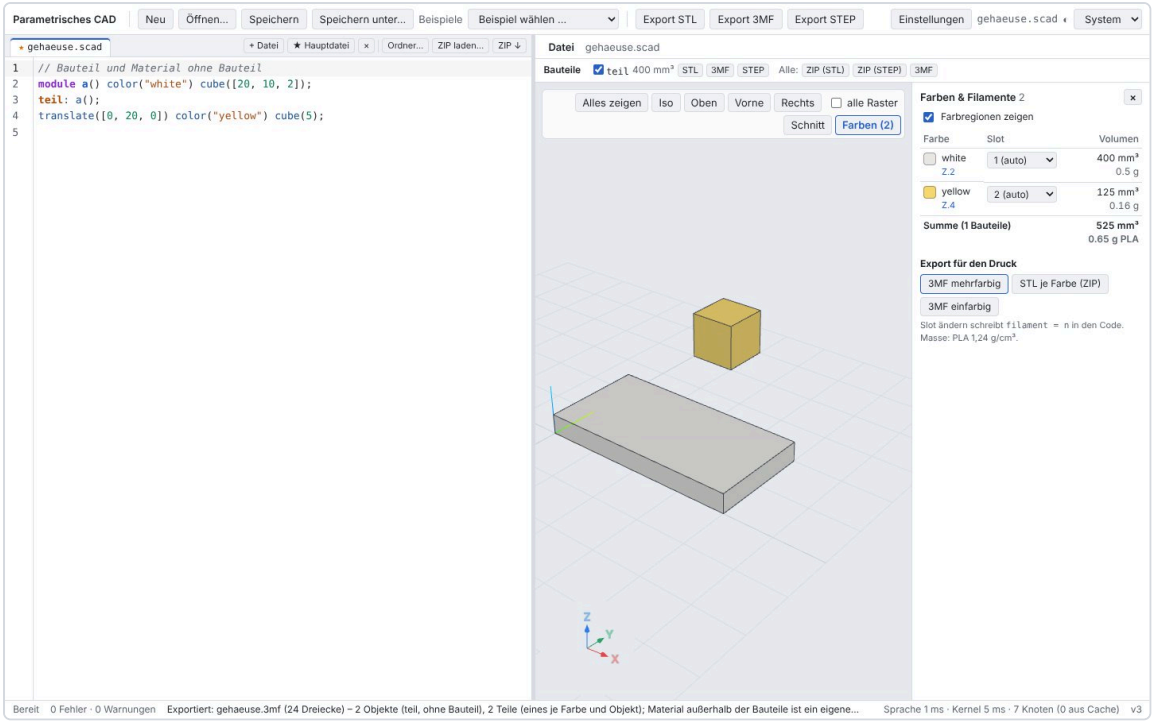


Extrusion with 2D color, warning line 3

- 48752 red pixels on plate and fillet
- overlap: 78063 blue > 21943 red pixels
- warning „Farben im 2D-Profil …“ in line 3, list default, red, blue

G04 F4: material without part, 3MF for Bambu Studio

A part teil and a yellow cube without part: before, 0 mm³ in the list and omitted in the UI export. Now 125 mm³, separate object „ohne Bauteil“, same file as cad export. The 3MF no longer has a color group: Bambu Studio opens no color dialog and takes the slots from model_settings.config.

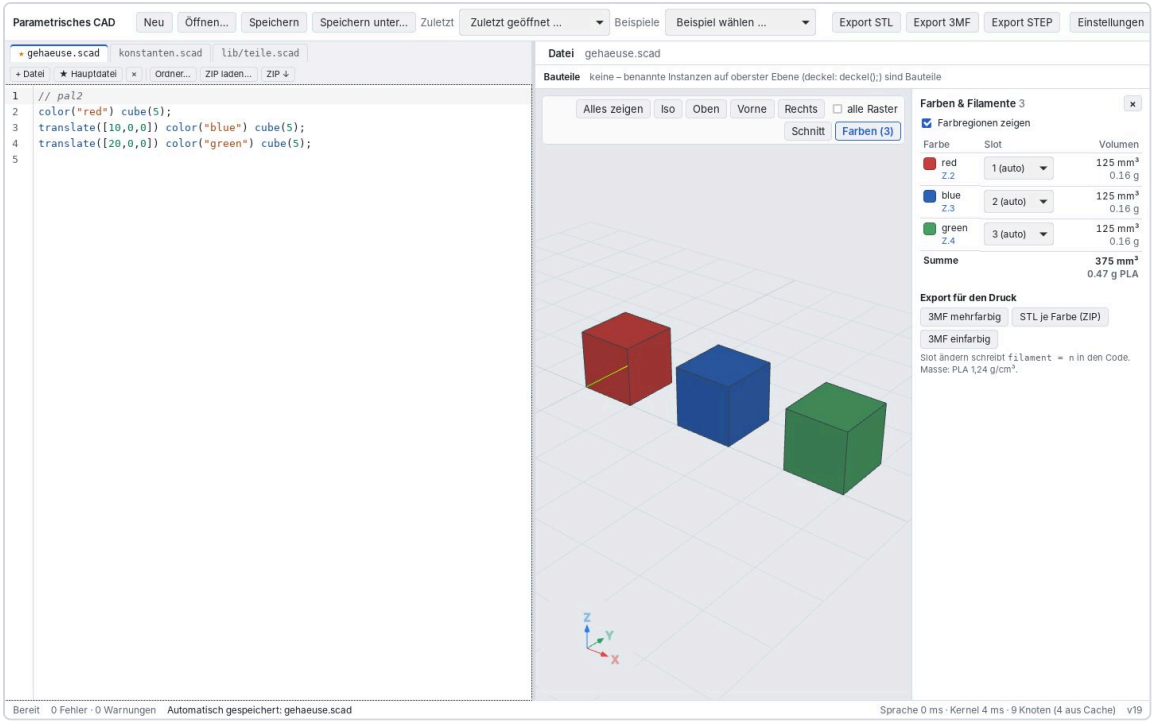


Export message with objects

- objects teil, „ohne Bauteil“
- no m:colorgroup, no m:color
- notes (translated): 2 objects (teil, „ohne Bauteil“), 2 sub-parts (one per color and object); material outside the parts is a separate object „ohne Bauteil“; filament slots in Metadata/model_settings.config, without color group: Bambu Studio and OrcaSlicer take over the slots without a color dialog

T15 (Tauri) Report 12 in Tauri: color regions per instance, filament() table, object „ohne Bauteil“, 3MF without color group

- colors red, blue, green
- no filament = 4 on color(c)
- slots red:4, blue:1, green:2
- objects teil, „ohne Bauteil“
- no color group
- notes (translated): 2 objects (teil, „ohne Bauteil“), 2 sub-parts (one per color and object); material outside the parts is a separate object „ohne Bauteil“; filament slots in Metadata/model_settings.config, without color group: Bambu Studio and OrcaSlicer take over the slots without a color dialog



pal2.scad: three identical cubes in three colors (color regions)

5 Test runs

Run	Result
`cargo test --workspace --no-fail-fast`	404 passed, 0 failed
`cargo clippy --workspace --all-targets -- -D warnings` · `cargo fmt --check`	clean · clean
`pnpm typecheck` · `pnpm test` (Vitest)	clean · 195 passed
`pnpm e2e` (Playwright, Chromium)	88 / 88
Tauri-E2E (`web/e2e-tauri/desktop.e2e.mjs`)	15 / 15
`tests/openscad-compat/compare.py`	33/33
`tests/step/verify.py`	48/48 passed
`tests/alltag/check.sh` (volumes of the everyday models)	19 models, 0 with problems, 175 s, exports in /tmp/alltag-d1zsigol (deleted)

New Rust tests:

- `long_ipc_dir_uses_short_socket_path` **ok**
- `f4_material_outside_parts` **ok**
- `f1_equal_geometry_keeps_its_colour_in_view_and_step` **ok**
- `f3_view_matches_the_print` **ok**
- `f6_no_slivers_between_same_coloured_neighbours` **ok**
- `f8_slot_remarks_and_single_colour` **ok**
- `f9_per_colour_export_and_list` **ok**
- `f7_value_rules` **ok**
- `f5_random_colours_keep_their_distance` **ok**
- `all_review_cases_evaluate_and_regions_add_up` **ok**
- `color_export::tests::no_color_group_by_default` **ok**
- `filament_declarations` **ok**
- `random_colours_do_not_collide_in_a_model` **ok**

- `filament_table_rules` **ok**
- `set_filament_of_a_parameter_colour_writes_a_table` **ok**
- `material_outside_parts_is_exported` **ok**

6 Limitations

- The view shows the exact regions only after they are computed (booleans per color node, cached); until then, while dragging and for the highlighted subtree under the cursor, the approximated color per face applies. If the cursor is on a statement, the rest is shown dimmed (as before).
- STEP does not split faces at color boundaries (a merged face gets the majority color).
- Mixed colors in a 2D profile below an extrusion are not separated (warning).
- Random colors that have to move away within the model depend on the names before them.
- Bambu Studio and OrcaSlicer were not checked in the programs themselves, but via their source code and a model of the importer (`tools/report/verify3mf.py`).
- The substitute socket is always under `/tmp` (not `$TMPDIR`, which can itself be long).